MLOPS Assignment 2

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1. Which run had the lowest latency, and why might that be?

- run_batch8 had the lowest latency, at 0.120 s.
- run_batch4 took 0.133 s, and run_batch1 was the slowest at 0.226 s.
- With batch size 8, the model processes 8 sentences together in one forward pass, so the 10 sentences need only 2 model calls.
- Fewer calls means less overhead per call, so the total time is lower.
- run_batch1 made 10 separate calls, and since it ran first it also paid a one-time warm-up cost.
- So latency decreases as batch size increases.

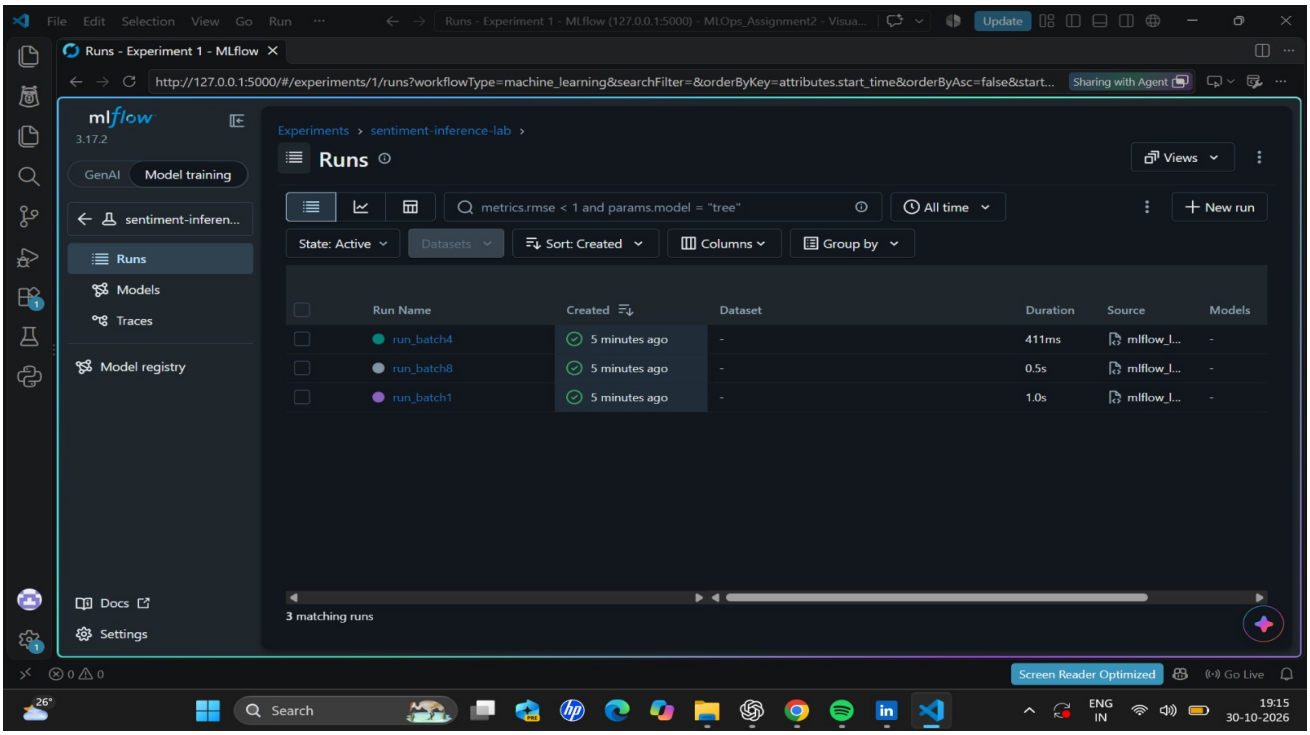
2. Did accuracy change between runs? Why or why not?

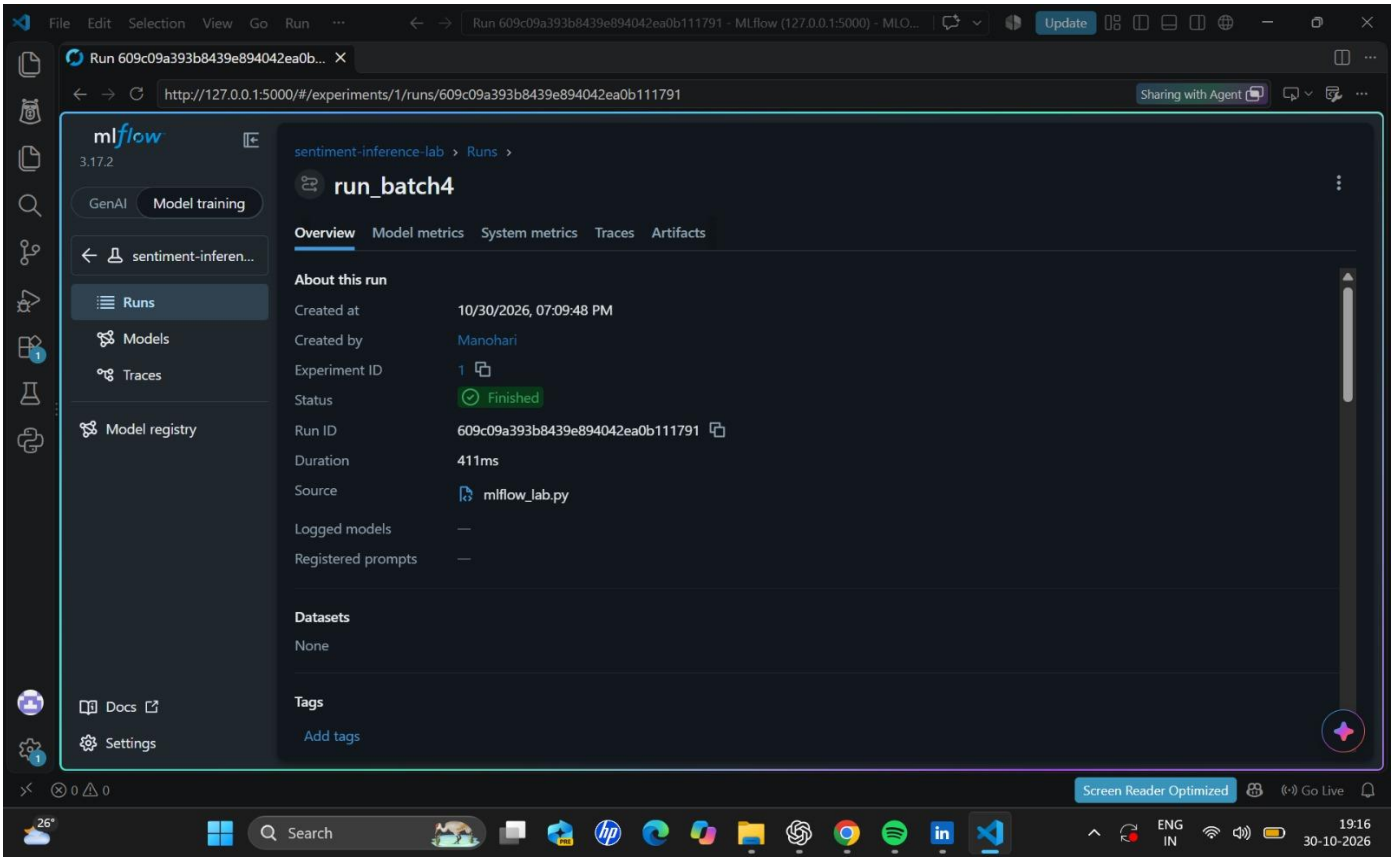
- No, accuracy stayed at 1.0 (100%) in all three runs.
  Every run used the same pre-trained DistilBERT model with the same weights.
- All runs were tested on the same 10 sentences.
- Inference is deterministic, meaning the same input always gives the same output.
- Batch size only changes how fast the sentences are processed, not the predictions.
- Average confidence was also the same (0.99965) in all runs.

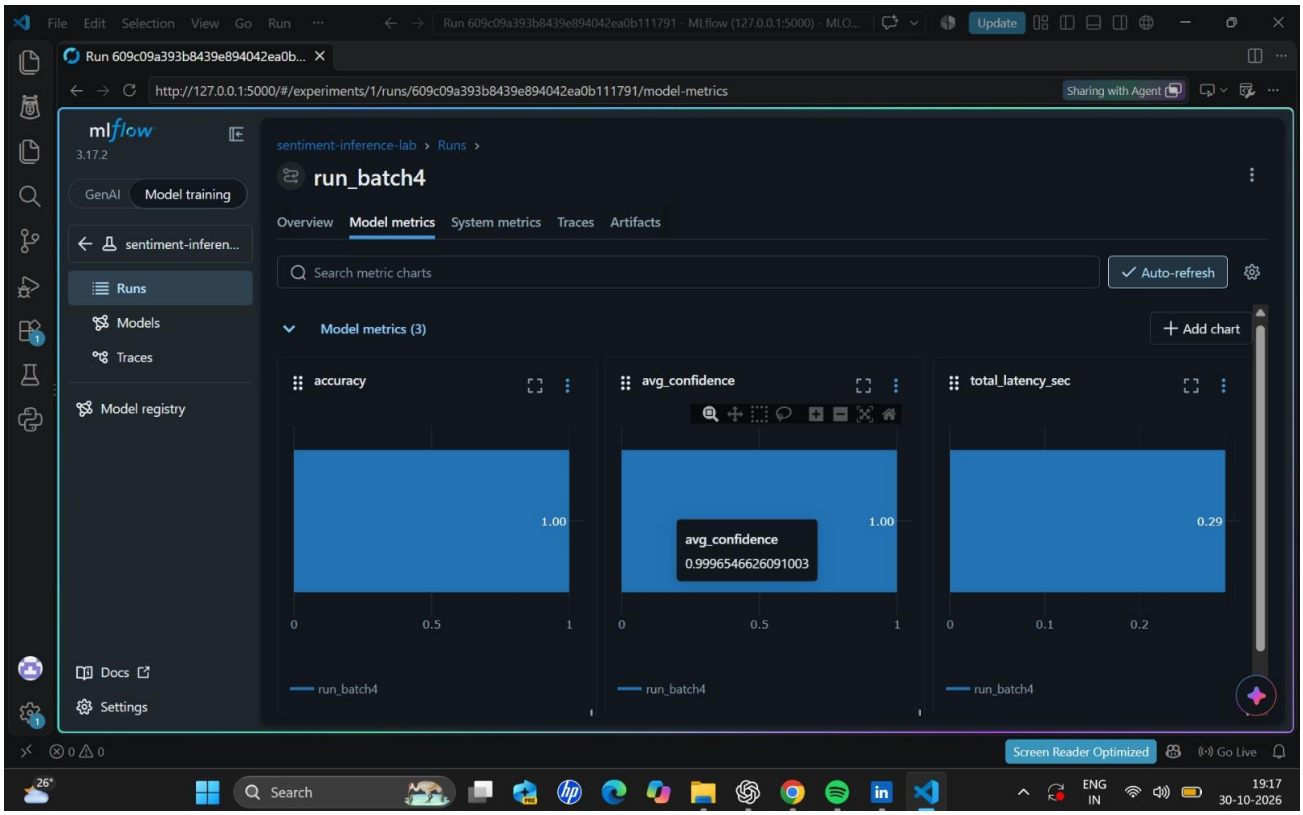
3. Name two benefits of using MLflow here instead of printing results in the terminal.

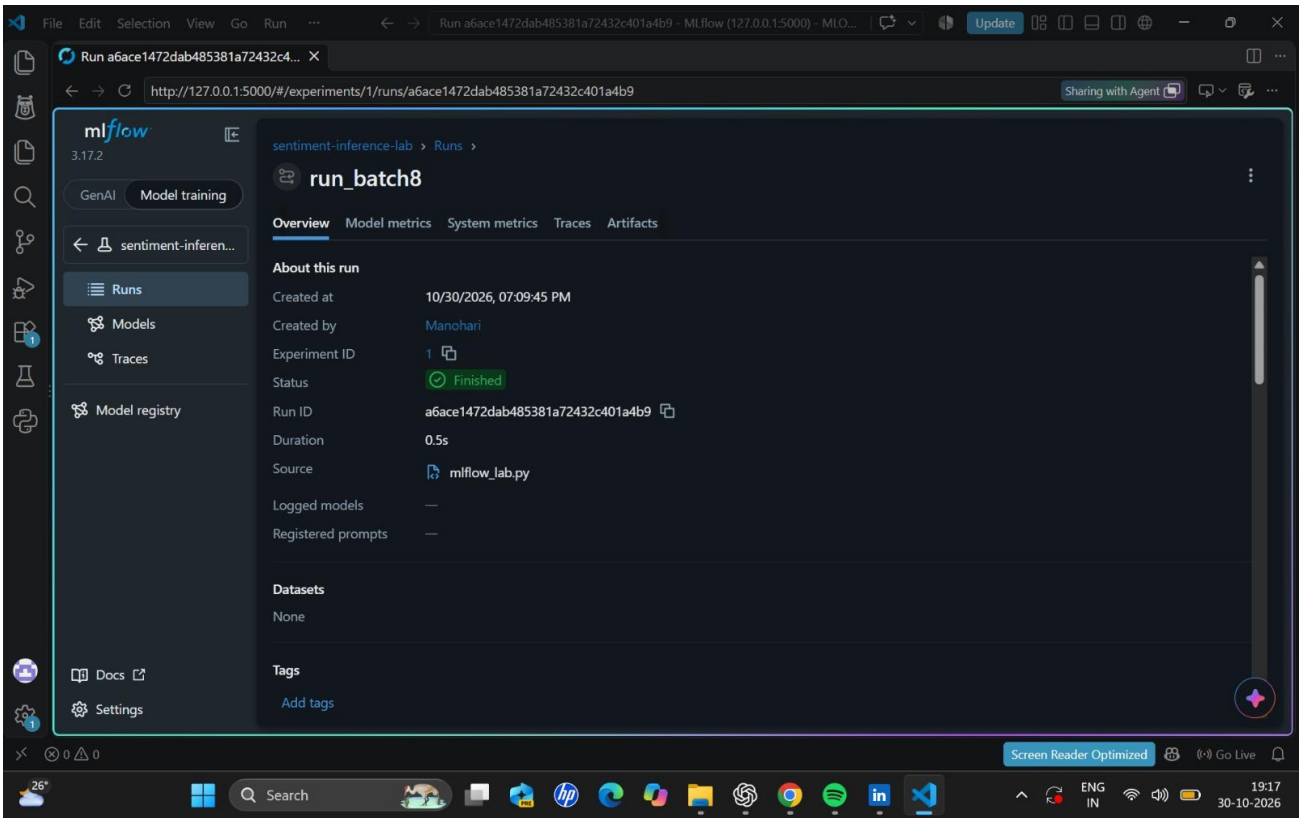
- Reproducibility and record-keeping: the parameters, metrics and artifacts (predictions.csv) of every run are saved permanently. Terminal output is lost once the window closes.
- Easy comparison: runs can be viewed and compared side by side in the MLflow UI, with tables and charts, instead of reading through printed lines.

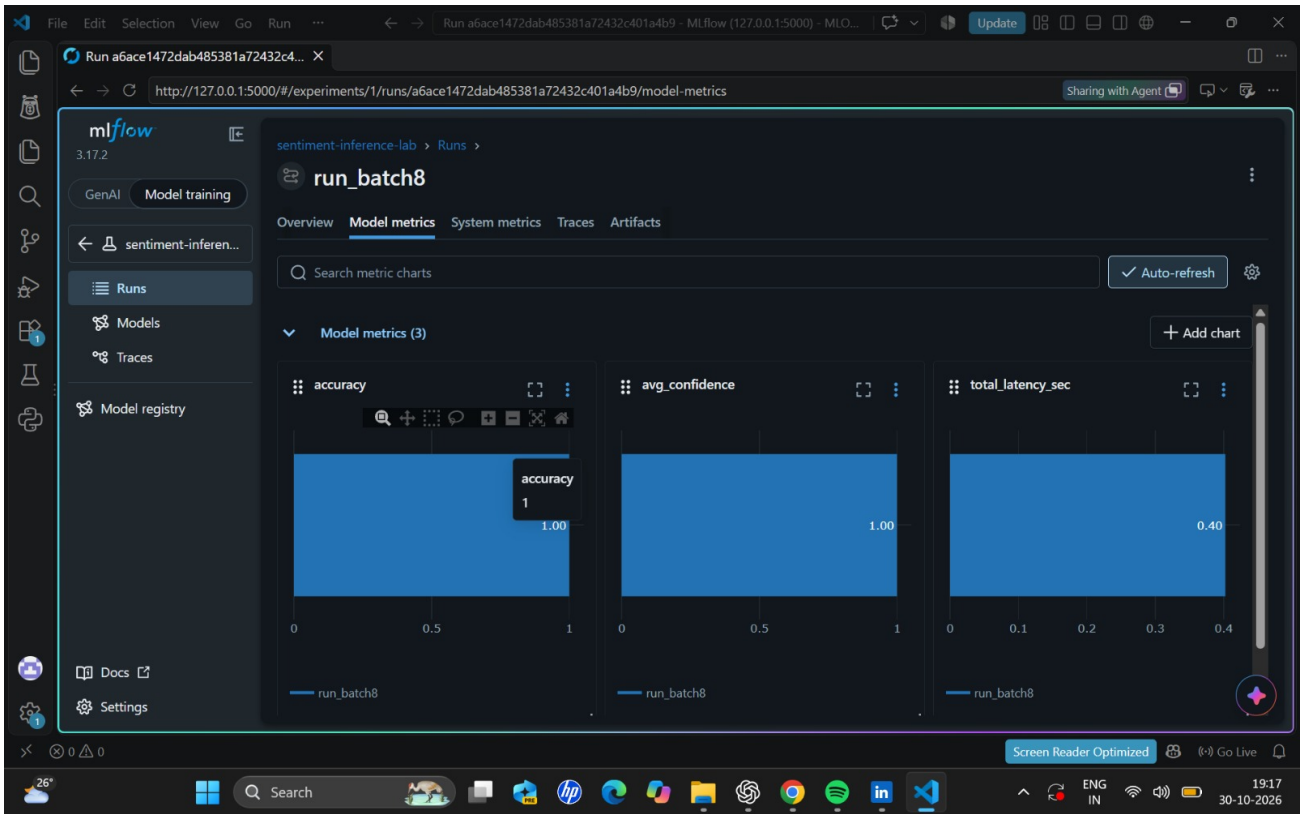
Screenshot of the MLflow runs table:

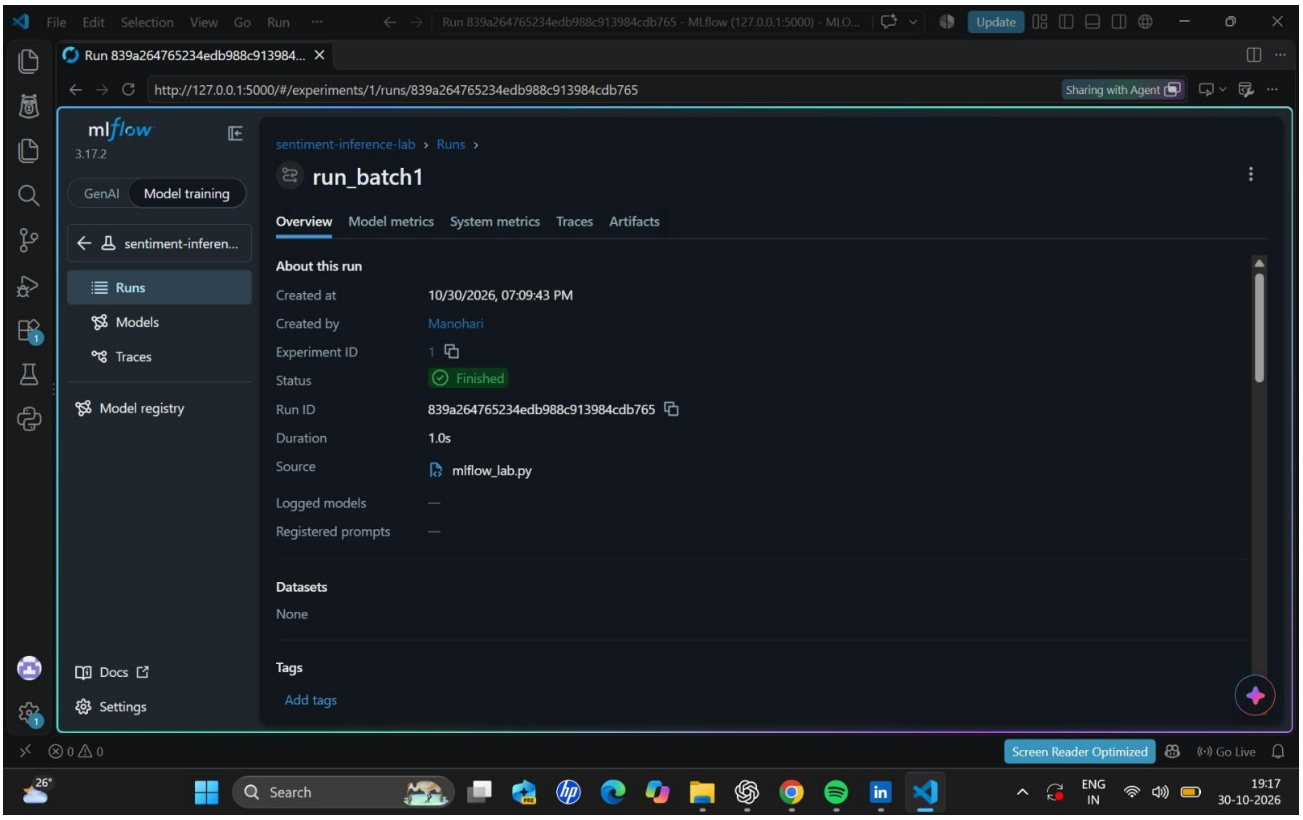


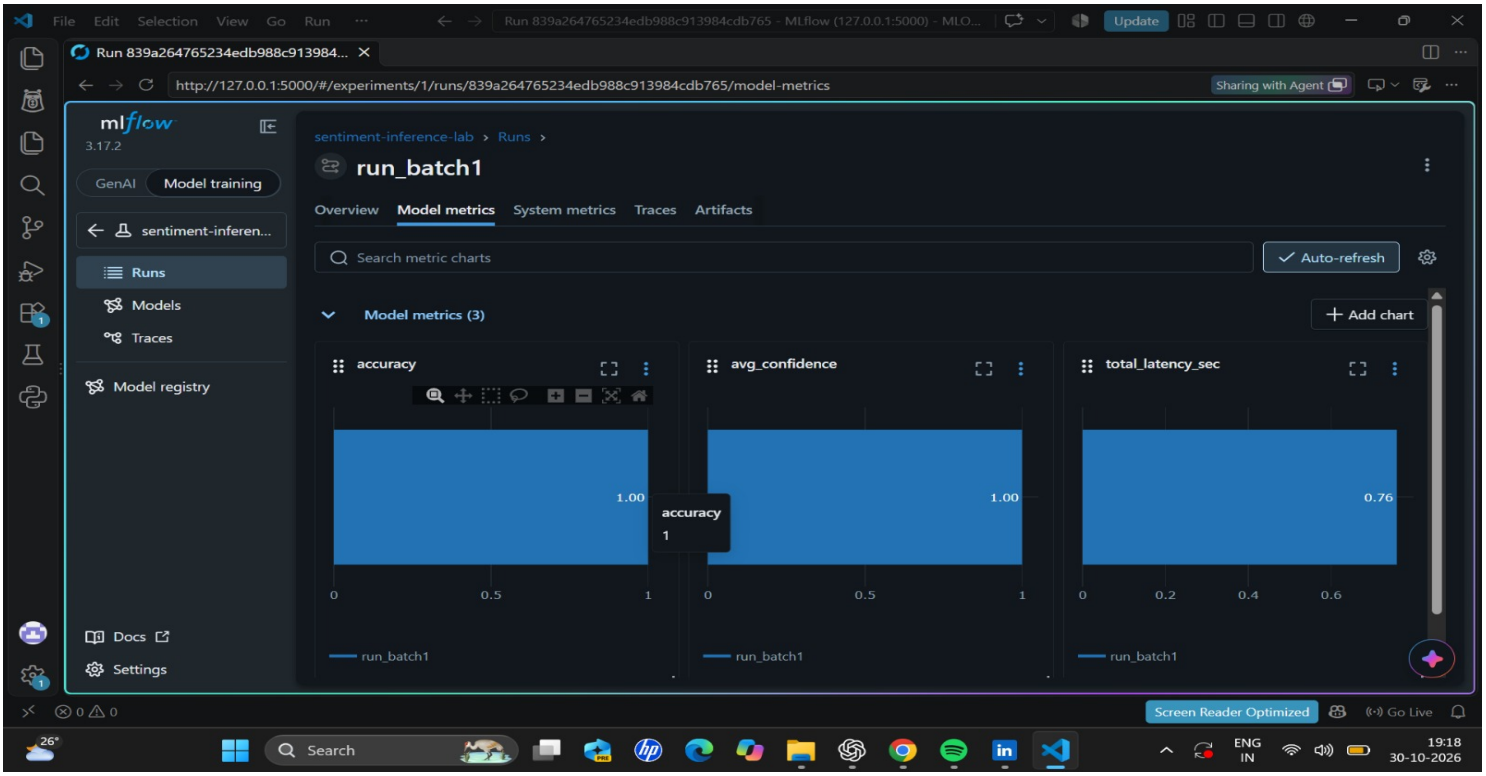



Run 609c09a393b8439e894042ea0b...
http://127.0.0.1:5000/#/experiments/1/runs/609c09a393b8439e894042ea0b111791/model-metrics
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch4
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
0.29
avg_confidence
0.9996546626091003
0
0.5
1
0.1
0.2
run_batch4
Screen Reader Optimized
Go Live
Search
ENG
IN
19:17
30-10-2026

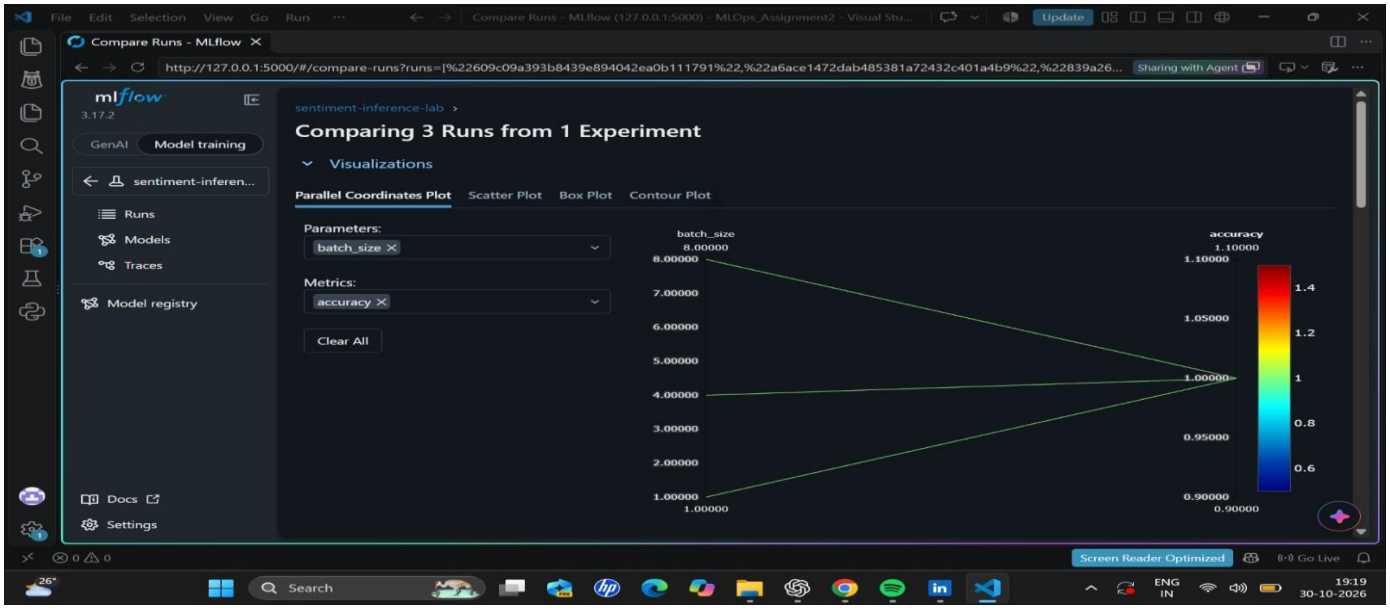

Run a6ace1472dab485381a72432c4...
http://127.0.0.1:5000/#/experiments/1/runs/a6ace1472dab485381a72432c401a4b9
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
About this run
Created at
10/30/2026, 07:09:45 PM
Created by
Manohari
Experiment ID
1
Status
Finished
Run ID
a6ace1472dab485381a72432c401a4b9
Duration
0.5s
Source
mlflow_lab.py
Logged models
—
Registered prompts
—
Datasets
None
Tags
Add tags
Screen Reader Optimized
Go Live
Search
ENG
IN
19:17
30-10-2026


Run a6ace1472dab485381a72432c401a4b9 - MLflow (127.0.0.1:5000) - MLO...
Update
Run a6ace1472dab485381a72432c4...
http://127.0.0.1:5000/#/experiments/1/runs/a6ace1472dab485381a72432c401a4b9/model-metrics
Sharing with Agent
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
accuracy
1
1.00
1.00
0.40
0
0.5
1
0.1
0.2
0.3
0.4
run_batch8
Screen Reader Optimized
Go Live
Search
ENG
IN
19:17
30-10-2026


Run 839a264765234edb988c913984cdb765 - MLflow (127.0.0.1:5000) - MLO...
Update
Run 839a264765234edb988c913984...
http://127.0.0.1:5000/#/experiments/1/runs/839a264765234edb988c913984cdb765
Sharing with Agent
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch1
Overview
Model metrics
System metrics
Traces
Artifacts
About this run
Created at
10/30/2026, 07:09:43 PM
Created by
Manohari
Experiment ID
1
Status
Finished
Run ID
839a264765234edb988c913984cdb765
Duration
1.0s
Source
mlflow_lab.py
Logged models
—
Registered prompts
—
Datasets
None
Tags
Add tags
Screen Reader Optimized
Go Live
Search
ENG
IN
19:17
30-10-2026



Screenshot of the Compare view:



mlflow_lab.py file:

